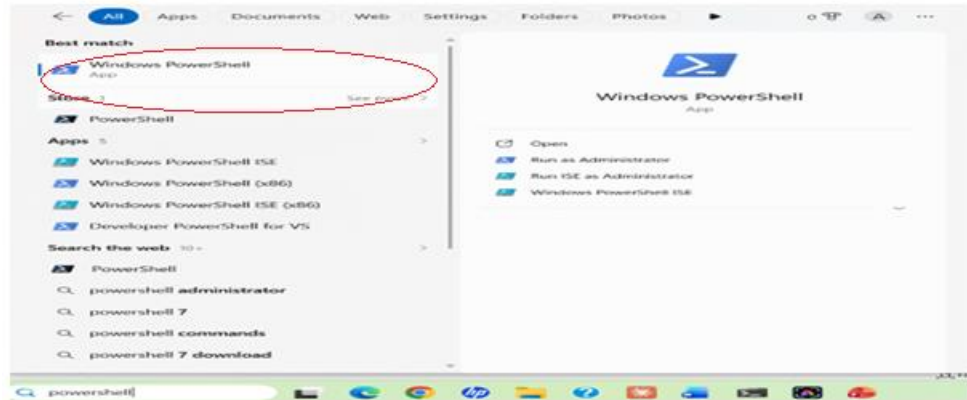
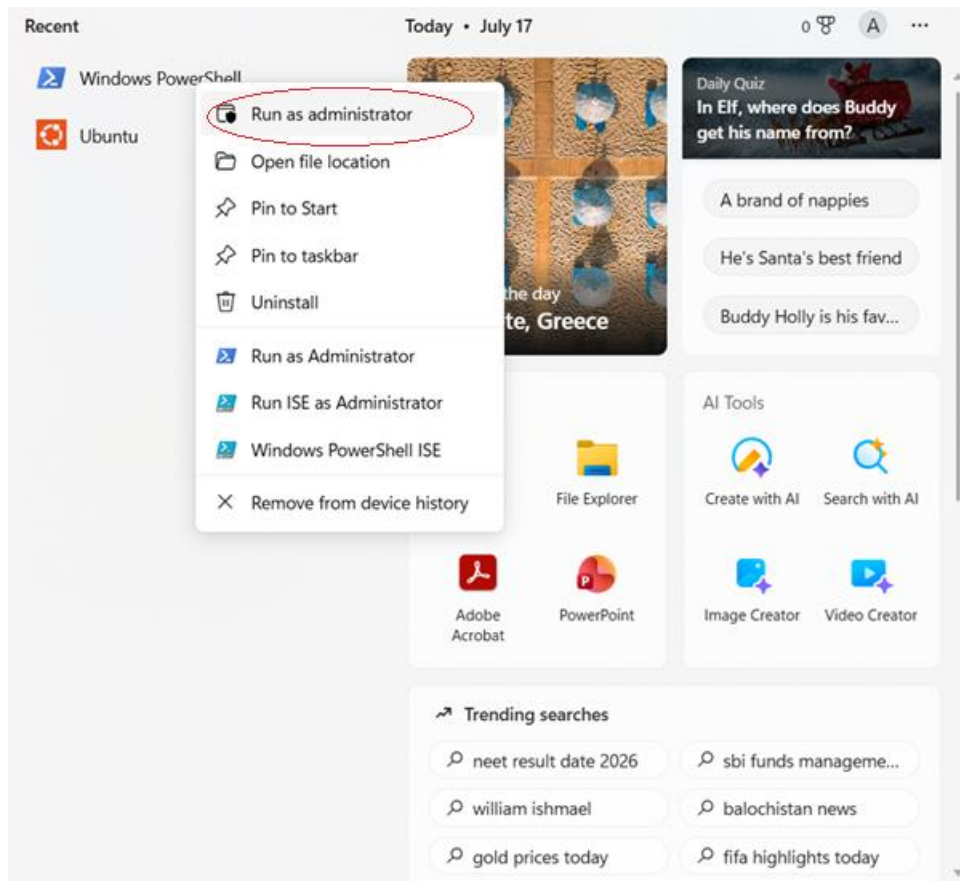


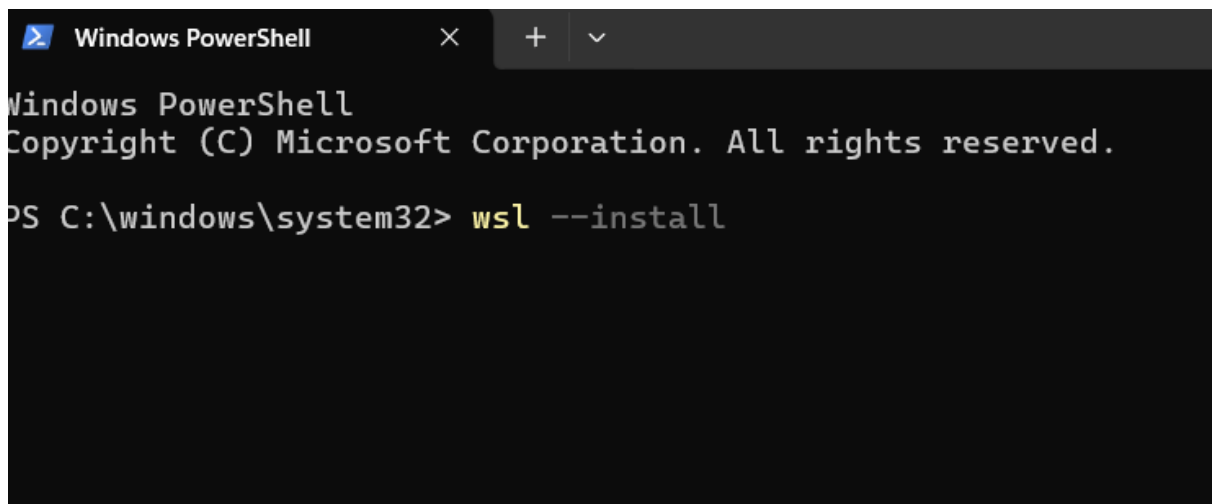
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', and 'PS C:\windows\system32> wsl --install'. The 'wsl' command is highlighted in yellow.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Fetched 3159 kB in 7s (470 kB/s)
34 packages can be upgraded. Run 'apt list --upgradable' to see them.
vedsk0831@DESKTOP-83FU7HB:~$ |
```

Step 7: Install GCC (C Compiler)

Step 8: Verify the Installation

```
vedsk0831@DESKTOP-83FU7HB:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR
PURPOSE.
```

Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program

```
vedsk0831@DESKTOP-83FU7HB:~$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.0
4.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 34
vedsk0831@DESKTOP-83FU7HB:~$
```

```
vedsk0831@DESKTOP-83FU7HB:~$ nano process.c
vedsk0831@DESKTOP-83FU7HB:~$ gcc process.c -o process
vedsk0831@DESKTOP-83FU7HB:~$ ./process
Enter a Linux command: pwd
```

Parent Process

Child Process

Parent PID : 5734

Child PID : 5735

Child PID : 5735

Parent PID: 5734

/home/vedsk0831

Child process completed.

```
vedsk0831@DESKTOP-83FU7HB:~$
```

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

The screenshot displays a GitHub repository interface. At the top, the repository name '2520030350-OPERATING-SYSTEM' is visible, along with the owner 'vedikakulkarni85'. Below this, the repository structure is shown, including two folders: 'OS practical programs' and 'OS skill programs', both of which were added via upload 2 weeks ago. The README section is currently empty, with a prompt to 'Add a README' and a note to help people understand the project. On the right side, the 'About' section is empty, and the 'Releases' and 'Packages' sections also show no published content.

Fork()

```
vedsk0831@DESKTOP-83FU7HB:~$ gcc fork.c -o fork
vedsk0831@DESKTOP-83FU7HB:~$ ./fork
Hello
Hello
vedsk0831@DESKTOP-83FU7HB:~$ |
```

Exec()

```
-rw-r--r-- 1 vedsk0831 vedsk0831 159 Jul 31 10:07 end.c
-rw-r--r-- 1 vedsk0831 vedsk0831 0 Jul 31 04:00 example.txt
-rw-r--r-- 1 vedsk0831 vedsk0831 264 Jul 31 10:09 exe.c
-rwxr-xr-x 1 vedsk0831 vedsk0831 15992 Aug 24 05:19 exex
-rw-r--r-- 1 vedsk0831 vedsk0831 174 Aug 24 05:19 exex.c
drwxr-xr-x 2 vedsk0831 vedsk0831 4096 Jul 31 03:58 folder
-rwxr-xr-x 1 vedsk0831 vedsk0831 15992 Aug 24 05:16 fork
-rw-r--r-- 1 vedsk0831 vedsk0831 104 Aug 24 05:16 fork.c
-rwxr-xr-x 1 vedsk0831 vedsk0831 15992 Aug 14 10:14 linux
-rw-r--r-- 1 vedsk0831 vedsk0831 163 Aug 14 10:14 linux.c
-rwxr-xr-x 1 vedsk0831 vedsk0831 16040 Aug 14 10:05 multiple
-rw-r--r-- 1 vedsk0831 vedsk0831 171 Aug 14 10:05 multiple.c
drwxr-xr-x 2 vedsk0831 vedsk0831 4096 Jul 31 03:58 my
drwxr-xr-x 2 vedsk0831 vedsk0831 4096 Jul 31 04:01 myfolder
-rw-r--r-- 1 vedsk0831 vedsk0831 78 Aug 4 03:44 myfolder.c
-rw-r--r-- 1 vedsk0831 vedsk0831 230 Jul 31 10:01 parentchild.
c
-rw-r--r-- 1 vedsk0831 vedsk0831 288 Jul 31 10:02 parentchildi
d.c
-rwxr-xr-x 1 vedsk0831 vedsk0831 16080 Aug 14 09:58 parentwait
-rw-r--r-- 1 vedsk0831 vedsk0831 264 Aug 14 09:58 parentwait.c
-rwxr-xr-x 1 vedsk0831 vedsk0831 16392 Aug 24 04:59 process
-rw-r--r-- 1 vedsk0831 vedsk0831 778 Aug 24 04:59 process.c
-rwxr-xr-x 1 vedsk0831 vedsk0831 16088 Aug 14 09:50 processid
-rw-r--r-- 1 vedsk0831 vedsk0831 112 Aug 14 09:43 processid.c
-rwxr-xr-x 1 vedsk0831 vedsk0831 16000 Aug 4 03:46 shell
-rw-r--r-- 1 vedsk0831 vedsk0831 110 Aug 4 03:46 vedika.c
drwxr-xr-x 2 vedsk0831 vedsk0831 4096 Jul 28 03:59 vedu
-rw-r--r-- 1 vedsk0831 vedsk0831 274 Jul 31 10:03 wait.c
-rw-r--r-- 1 vedsk0831 vedsk0831 159 Jul 31 09:55 week1.c
vedsk0831@DESKTOP-83FU7HB:~$ |
```